

AI TRAINING – ASSIGNMENT 01

AI Fundamentals: Explore, Understand & Explain

Self-Exploration Assignment | Beginner Level

Topic Area	AI overview, AI vs ML vs DL vs GenAI, use cases, and career map
Mode	Self-exploration + research + practical thinking
Suggested Time	2–3 hours
Submission	One document / PDF containing your answers, examples, and reflections

1. What is this assignment about?

This is not a memorization assignment. Your goal is to explore AI yourself, understand the basic concepts, and explain them in simple words. You are expected to search, watch short videos if needed, use AI tools for clarification, and compare information from multiple sources.

Important: Do not simply copy an AI-generated answer. Use AI as a learning assistant, then rewrite the ideas in your own words and add your own examples.

2. Learning Goals

- Understand what Artificial Intelligence means at a high level.
- Clearly distinguish AI, Machine Learning, Deep Learning, and Generative AI.
- Identify real-world AI use cases across different industries.
- Understand the major career paths available in AI and related technology roles.
- Explain the concepts in your own words instead of relying on textbook definitions.
- Apply the concepts to a practical real-world example.
- Identify common misunderstandings or errors and improve your explanation.

3. Your Mission

Imagine that a friend who knows nothing about AI asks you: “What exactly is AI, what is the difference between AI, ML, DL and GenAI, where is AI used, and what can I become if I learn AI?”

Your job is to create a beginner-friendly answer that your friend can understand in 10 minutes.

4. Step 1 – Explore: What is AI?

Do your own research and answer the following questions:

1. What is Artificial Intelligence in your own words?

Ans: AI is a technology that is used in computer systems that to perform tasks which require human intelligence. AI can reduce our time doing same task again and again. By using AI we can solve very complex problems. AI is trained by using data. For example when we saw a dog on that road we can simply say it is dog. Because we saw many dogs before. Our mind will store shape of the dog. AI is also same like it observes the image and identifies the pattern and stores it. When ever it gets same or similar input it will give answer.

2. What makes a system 'intelligent'?

Ans: A system is said to be intelligent when it performs the tasks without interaction of human. And performs tasks on its own like when ever user gave an input it must identify it and provide and also it must be self learn by using data and provide decisions correctly.

3. Give at least 5 examples of AI that you may interact with in daily life.

Ans: Face recognition, chat bots, email spam detection, Instagram recommendations, Google Maps.

4. Choose one AI example and explain what the system is doing.

Ans: Let us take example as email spam detection. If we did not use AI and made manual rules like if mail contains words like rewards or any suspicious content sent to spam. What if the mail contains no words that you mention it is in useful content. This can be overcome by using AI. AI first identifies all mails and identifies the patterns and content of spam mails and non spam mails. It completes the analysing the mails then it will identify the mails which is spam or not by using already identified patterns. It will evaluate the mail based on percentage like "Your bank account will be blocked today. Click here immediately to verify our account." This type of emails human cannot identify as spam but AI can.

5. Is every software application an AI application? Explain why or why not.

Ans: No all the software applications are AI applications because we can see some applications like calculator which is predefined by mathematical rules there we don't use AI. There are some tasks that doesn't require AI to perform. It just needs some basic rules.

6. What are 2 things AI can do well and 2 things AI may struggle with?

Ans: AI can do repetitive work many without getting tired and can identify the pattern in large amount of data. The things that AI will struggle is when it gets input which it does not know about that means let us take example of automated cars when it enters into the which is new and unique. AI can understand the emotions of human but cannot feel it.

Research rule: Use at least 3 learning sources. These can be articles, videos, official documentation, or educational websites.

Tip: Try searching for simple explanations first. If two sources disagree, investigate why.

5. Step 2 – The Big Picture: AI vs ML vs DL vs GenAI

This is the most important part of the assignment. Do not memorize definitions. Build a mental model.

First, complete this comparison table in your own words:

Concept	What is it?	How does it learn/work?	Simple example	Relationship	Your analogy
AI	Ai is technology that used by computer systems and performs the tasks without human interaction.	The workflow of Ai first it takes the data and analyze it and find the patterns in it and it will learn from that patterns and made decisions . it only provide output from what it have learned from.	Mail detection,yout ube recommendati ons	Ai is large field that include ml and dl	In my words ai is like smart assistant which helps us to save time by doing taks fatser
Machine Learning (ML)	ML stands for machine learning .it is about training machines by using data instead of writing programming for every action.In machine learning models should be train by using many types of data like supervised learning, unsupervised learning and semi supervised learning.	In machine learning the model is trains with data. after training it predicts and suggest the output. If the trained data is inaccurate then the output is also inaccurate. The process of training the model is at first collect the data and clean it. train the model with the data and then finally predict the outputs.	Crop diseases detection and spam mail detection	Ai made intelligent machines ML train the machines by using the data.	ML is nothing training the machines by using the data.it also learn from the experience.
Deep Learning (DL)	DL is also like ml .but DL is	The usage of DL is like	face recognition	DL is almost like ML but it	to solve the complex or

	used to solve the complex problems by analyzing the input with somany layers .this used the artificial neural networks which works like a human brain.	ML .but in ml it learns from the raw data and labels.but in DL the input is taken and hidden layers focus on the each part of the input like corners and contours.by analyzing the input so clearly and providing the ouput. It can identify the patterns in many complex problems which humans cannot solve.	and leaf diseas detection	evaluates the inputs much deeply and produces the output accurately.	gets accurate output we can use the DL. But it takes large amount of data to train.
Generative AI (GenAI)	Generative ai is ai that learns patterns from daa and generates the new content like images ,audio and vidoes.	gen ai collect the data and learns the patterns from the data and takes input as prompt and generates the new content based on the input.	Generates the audios ,images ,videos	It is a ai system that generates the new content.	It is majorly used for the generating the new content like audios and images by using the prompt.

Now answer these questions:

1. Is ML a part of AI? Explain with a simple example.
2. Is Deep Learning a type of Machine Learning? Explain.
3. Is Generative AI the same thing as AI? Why?
4. Can an AI system exist without Machine Learning? Research and explain one example.
5. Draw a simple diagram showing the relationship between AI, ML, DL and GenAI.
6. Create one real-world analogy to explain all four concepts together.

6. Step 3 – Use Case Hunt

Find AI applications in different industries. Do not use only ChatGPT examples.

Industry	AI use case	AI / ML / DL / GenAI?	Who benefits?	What could go wrong?
Banking / Finance	Fraud	Machine Learning .	Banks, customers.	sometimes correct transactions also named as fraud.
Healthcare	Disease detection	Deep Learning.	Doctors, patients.	Sometimes AI may give an incorrect diseases detections.
Education	students can learn using ai.	AI / Machine Learning	Students, teachers.	AI may provide incorrect answers.
Retail / E-commerce	Product recommendations	Machine Learning .	Customers.	AI may collect too much user data.
Manufacturing	Research + write	Research + write	Research + write	Research + write
Agriculture	Crop disease detection	Deep Learning .	Farmers.	AI may incorrectly identify a crop disease.
Transportation	Traffic prediction	Machine Learning .	Drivers, passengers.	Wrong routes predictions.
Entertainment / Media	Content recommendations	Machine Learning / GenAI	Viewers, content creators.	AI may recommend inaccurate content.

Challenge: Find one AI use case that you think is surprisingly useful and explain why.

7. Step 4 – Mini Practical Exercise: Design an AI Solution

Choose ONE problem from your college, personal life, local community, or a business you know. Design a simple AI-based solution. You do not need to build the software.

1. What problem are you trying to solve?

Ans: I am trying solve which farmers are continuously getting that problem that diseases detection by using machine learning and deep learning .

2. Who faces this problem?

Ans: Majorly this problem was faced by the farmers

3. What information/data would your system need?

Ans: For detecting the diseases in crops it requires the images of the crops to detect what type of diseases occurred

4. Would you use AI, ML, DL, GenAI, or a combination? Why?

Ans: I would use both ml and dl because I want to train my model more and more times to get accurate values. i am using dl because it will evaluate the images very accurately by using neural networks .it will check every pinpoint of the image by every edges and contours of the image and providing the best accurate solutions.

5. What would the system take as input?

Ans: The should take the image of the crop as and produce the output is the crop is infected or not.

6. What would the system produce as output?

Ans: It provide output as is the crop id infected or not.

7. Give one example of how a user would interact with it.

Ans: User need to take the images of crops because we don't when the disease will attack the plant. that's why we need to take the images and upload in the app to get results. then farmer may protect his field before the disease spread the whole the field need to monitor the field every day.

8. What could go wrong with your solution?

Ans: If my solution has gone wrong the I need to train my model with more crops and with more images. then it will some accurate of producing solutions

9. How would you reduce or handle that risk?

Example idea (do not copy): A college assistant that answers student questions about assignments, attendance rules, and course material.

8. Step 5 – Career Map: Where Can AI Take You?

Explore AI-related career paths. The goal is not to decide your career today; it is to understand what different roles actually do and what skills they require.

Role	What do they do?	Key skills	Typical tools/technologies	Would I explore it? Why?
AI / ML Engineer	Build and train AI/ML models to solve real-world problems.	Python, Machine Learning, Deep Learning, Mathematics.	Python, TensorFlow, PyTorch, Scikit-learn.	Yes, I am intrest in creating ai systems.
Data Scientist	used to find the patterns	Python, Statistics, Machine Learning, Data Analysis.	Python, Pandas, NumPy, Jupyter, SQL.	Yes, because I like working with data and finding

				patterns.
Data Analyst	Analyze data and create reports used understand information.	Excel, SQL, Data Analysis, Communication.	Excel, SQL, Power BI, Tableau, Python.	Yes, because it is a good starting role for learning data.
Generative AI / LLM Engineer	Build applications using AI models that can generate text, images, code, and other content.	Python, LLMs, Prompt Engineering, APIs.	Python, OpenAI APIs, Hugging Face, LangChain.	Yes, because Generative AI is useful for building modern applications.
AI Product Manager	Plan and manage AI products from idea to launch.	Communication, Product Management, AI knowledge, Problem-solving.	Jira, Trello, Notion, Analytics tools.	Yes, because I am interested in combining technology and product development.
AI Researcher	Research new AI methods and develop improved algorithms and models.	Mathematics, Programming, Machine Learning, Research.	Python, PyTorch, TensorFlow, Jupyter.	Yes, because I want to understand how new AI technologies are developed.
Computer Vision Engineer	Research + write	Research + write	Research + write	Research + write
NLP Engineer	Research + write	Research + write	Research + write	Research + write
MLOps / AI Platform Engineer	Deploy, monitor, and maintain AI/ML models in production.	Python, Cloud, DevOps, Machine Learning.	Docker, Kubernetes, AWS, Git, MLflow.	Yes, because I want to learn how AI models are deployed and maintained.
AI Application Developer	Build applications that use AI models and AI APIs.	Python, HTML, CSS, JavaScript, APIs.	Python, JavaScript, REST APIs, OpenAI APIs, Git.	Yes, because it matches my interest in developing applications using AI.

9. Step 6 – Career Self-Check

Based on your exploration, answer:

Which AI-related role interests you most right now?

Ans: I am intrested in two Ai related roles those are Ai/ML engineer and Ai application developer.

Why does it interest you?

Ans: Because I am interested in developing many apps and websites using AI and also interested in creating AI systems.

What skills do you already have that may help?

Ans: I had already learn basics which required for AI.

What are 3 skills you need to learn next?

Ans: I want to python libraries like Numpy, pandas . I also want to learn ML,DL.

What would your 2-month learning roadmap look like?

Ans: In month 1 I would like to learn python, and python libraries and machine learning basics and build 1 ml project. In month 2 I learn DL ,NN and gen AI also RAG basics build 1 AI and genAI project.

What would you like to build after learning the basics?

Ans: After learning basics I would like to build plant disease detection and study assistant.

10. Step 7 – Common Errors: Find and Fix Them

For each statement below, decide whether it is correct, partially correct, or incorrect. Then explain why.

1. "AI and Machine Learning are exactly the same thing."

Your verdict + correction: _____

NO both AI and ML are exactly same AI gives intelligent machines but ML trains the machines using the data.

2. "Deep Learning is unrelated to Machine Learning."

Your verdict + correction: _____

No there is relation between machine learning and deep learning in both the main aim is to train the machine with data. But in different ways.

3. "ChatGPT is just a search engine."

Your verdict + correction: _____

No ChatGPT is not only used searching it used for creating new content like audio, images and videos.

4. "Generative AI always gives correct answers."

Your verdict + correction: _____

no AI system gives the correct .the output is based on the data it is having.

5. "If a company uses AI, it does not need humans anymore."

Your verdict + correction: _____

No human power is require even the company uses the Ai. Because Ai does not have mind to think it always take decisions from the data and previous experiences .human need to control the Ai.

6. "More data automatically means a better AI system."

Your verdict + correction: _____

Yes if the Ai system trains using more data that means it gives more accurate outputs and predictions.

11. Step 8 – Teach It Back

Pretend you are the trainer. Explain the following in a way that a Class 10 student can understand:

1. What is AI?:

Ans: Ai is a technology that is used in computer systems that to perform tasks which require human intelligence.ai can reduce our time doing same task again and again. By using ai we can very complex problems .ai is trained by using data

2. AI vs ML:

Ans: Ai is used to create machines to perform tasks that require human intelligence. ML is subtopic of AI. ML is used to train that machines based on data it had

3. ML vs DL:

Ans: Machine learning is process of training the machines by using the data. Machines will train by using various types of learning methods like supervised learning, unsupervised learning and semi supervised learning. DL is same as ML in DL also used train machines by using data. but in DL I will there are three stages .at first it takes input after that in hidden layers .each layer will focus on each part if the input like corners and contours. ML is used for predictions and DL is used to solve the complex problems that humans unable to solve.

4. What is Generative AI?:

Ans: Generative Ai nothing but creating new content based patterns learned from data which is having in the machine.it is used to create new content like vidoes, audios, images.

5. One real-world AI use case:

Ans: Fraud transaction detection: Ai will analyse all the transaction if there are any transaction which looks very different it will flagged as fraud.it helps to prevent financial fraud.

6. One AI career path:

Ans: I had interest in ai application developer. Because I can develop any websites and apps by using Ai within a minutes. without wasting time on writing coding.

12. Final Challenge – 5-Minute AI Explanation

Record a 3–5 minute video or prepare a 1-page explanation as if you are teaching a friend. You must cover: AI → ML → DL → GenAI → real-world use case → career path.

Optional: Use a diagram, analogy, or story. Avoid reading a copied definition.

13. What to Submit

- AI explanation in your own words
- AI vs ML vs DL vs GenAI comparison table
- Relationship diagram
- 8-industry use-case table
- One AI solution design
- AI career map
- Career self-check
- Common errors and corrected explanations
- Teach-it-back answers
- Final 3–5 minute explanation OR 1-page summary

14. Self-Evaluation Checklist

I can...	Yes	Need More Practice	Notes
Explain AI without using a memorized definition.	☐	☐	
Explain the relationship between AI, ML, DL and GenAI.	☐	☐	
Give real-world examples of AI.	☐	☐	
Identify whether a use case involves ML, DL or GenAI.	☐	☐	
Explain at least 5 AI career roles.	☐	☐	
Design a basic AI solution for a real problem.	☐	☐	
Identify common AI misconceptions.	☐	☐	
Teach the basics of AI to another person.	☐	☐	

15. Rules for Using AI During This Assignment

- You MAY use ChatGPT, Gemini, Claude, Perplexity, YouTube, Google, and other learning resources.
- Use AI to explain, compare, question, and give examples—not to blindly copy your submission.
- If you use an AI-generated explanation, verify it using another source.
- Your final answers should be written in your own words.
- If you discover that an AI tool gave you an incorrect answer, mention the mistake and explain the correction. This is part of the learning.
- Add sources/links for important information you researched.

16. Trainer's Evaluation Guide

Area	What to look for	Suggested Weight
Concept understanding	Correct distinction between AI, ML, DL and GenAI	25%
Own explanation	Simple language, examples, analogies, not copied definitions	20%
Use cases	Relevant and correctly classified examples	15%
Practical thinking	Problem → data → AI approach → output → risks	20%
Career exploration	Understanding of roles and	10%

	skills	
Reflection & error correction	Can identify and fix misconceptions	10%

Remember: The goal is not to know everything about AI today. The goal is to learn how to explore, question, understand, and explain AI.